

Real-Time Fractional Tracking (R-TFT): Recursive Dimensionality Detection via Golden Ratio Locking

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Abstract

We reformulate dimensionality detection in the R-TFT framework using a recursive φ -locking mechanism. Instead of scanning for emergent attractors, we demonstrate that coherent dimensional structures naturally align with golden-ratio embeddings. Each dimensional candidate is evaluated by projecting real-time angular velocities onto φ -structured vectors, then scored for recursive coherence using φ -deviation metrics. The result is a detection scheme where dimensionality is not inferred statistically, but confirmed through resonance persistence and recursive minimality. This model bridges cosmological, quantum, and cognitive domains under a unified dimensional resonance principle.

1 Introduction

Traditional dimensionality detection methods rely on empirical feature counts or statistical embeddings. In contrast, R-TFT models dimensionality as a recursive signature: dimensions emerge when recursive phase-tracking vectors stabilize under golden-ratio alignment.

This paper replaces simulation-based attractor scanning with an intrinsic coherence test rooted in φ -recursive mathematics.

2 Theoretical Framework

Given a candidate dimension $d \in \mathbb{N}$, define a resonance hypothesis vector:

$$P_d = \underbrace{[\varphi, \varphi, \dots, \varphi]}_{d \text{ times}} \in \mathbb{R}^d \quad (1)$$

Compute its norm:

$$\|P_d\| = \sqrt{d \cdot \varphi^2} = \varphi\sqrt{d} \quad (2)$$

Then compute its φ -dimensional embedding:

$$\text{Dim}(d) = \frac{\ln \|P_d\|}{\ln \varphi} = 1 + \frac{\ln \sqrt{d}}{\ln \varphi} \quad (3)$$

And its coherence deviation:

$$C_\varphi(d) = |\text{Dim}(d) - \text{round}(\text{Dim}(d))| \quad (4)$$

3 Detection Criterion

A dimension d is deemed φ -stable if:

$$C_\varphi(d) < \varphi^{-1} \approx 0.618 \quad (5)$$

This identifies recursive coherence at dimensions whose embedding lies near a stable recursive orbit.

4 Cross-Domain Applications

We apply this method to several physical and informational domains:

- **CMB Resonance:** Low- ℓ multipoles align with φ -stable embeddings.
- **Quantum Systems:** Entangled states persist longer at recursively valid dimensional partitions.
- **Cognitive Models:** Working memory appears to stabilize near $\text{Dim} = 7.4$, a φ -projected band.

5 Conclusion

Rather than deriving φ as an emergent attractor, this framework treats it as the foundational basis of dimensional coherence. Recursive systems lock into dimensionalities that minimize C_φ , favoring persistence, memory, and cross-domain stability. Dimensionality is not merely a count, but a resonance.

Ethical Statement

This framework complies with the Resonance Ethics License (REL-2.0). It must not be used for surveillance, predictive coercion, or dimensional profiling of human behavior. Dimensional coherence detection must remain domain-agnostic and free of imposed behavioral control systems.

<https://github.com/qcfrag/Real-Time-Fractional-Tracking-R-TFT/blob/main/LICENSE.txt>